

INITIATE-PAH (CTP-PAH-001)

Anonymized Training Protocol — For Educational Use Only

Version 2.0 (Amendment 1) — Date: 2025-09-07

This protocol is a de-identified training case. It is inspired by publicly available clinical trial materials but contains no sponsor names, brand names, or proprietary content. All drug/product names are placeholders. Do **not** use clinically.

LIST OF ABBREVIATIONS AND ACRONYMS

• 6MWD	• EMA	Interactive Response Technology
6-minute walk distance	European Medicines Agency	• IP
• 6MWT	• EOS	Investigational product
6-minute walk test	End of study	• ISF
• AE	• EOT	Investigator Site File
Adverse event	End of treatment	• LVEDP
• ALT	• ERA	Left ventricular end-diastolic pressure
Alanine aminotransferase	Endothelin receptor antagonist	• MedDRA
• ANCOVA	• FAS	Medical Dictionary for Regulatory Activities
Analysis of covariance	Full Analysis Set	• m/d/sPAP
• AST	• FC	Mean/diastolic/systolic pulmonary artery pressure
Aspartate aminotransferase	Functional class	• mRAP
• ATS	• FDA	Mean right atrial pressure
American Thoracic Society	Food and Drug Administration	• MTD
• BMI	• FEV1	Maximum tolerated dose
Body mass index	Forced expiratory volume in 1 second	• NAION
• BP	• FVC	Non-arteritic ischemic optic neuropathy
Blood pressure	Forced vital capacity	
• CEC	• GCP	• NT-proBNP
Clinical Events Committee	Good Clinical Practice	N-terminal pro B-type natriuretic peptide
• CFR	• GMR	• PAH
Code of Federal Regulations	Geometric mean ratio	Pulmonary arterial hypertension
• cGMP	• HIV	• PAWP
Cyclic guanosine monophosphate	Human immunodeficiency virus	Pulmonary arterial wedge pressure
• CI	• HR	• PDE-5
Confidence interval	Heart rate	Phosphodiesterase-5
• CL		• PDE-5i
Confidence limit		Phosphodiesterase-5 inhibitor
• CO	• IB	• PH
Cardiac output	Investigator's Brochure	Pulmonary hypertension
• CRA	• ICF	• PAWP
Clinical Research Associate	Informed Consent Form	Pulmonary arterial wedge pressure
• CRF	• ICH	• PDE-5
Case Report Form	International Council for Harmonisation	Phosphodiesterase-5 inhibitor
• CRO	• IDMC	• PI
Contract Research Organization	Independent Data Monitoring Committee	Principal investigator
• CYP3A4	• IEC	• PPS
Cytochrome P450 3A4	Independent Ethics Committee	Per-protocol Set
• DLCO	• ILSDRB	• PVOD
Diffusing capacity of the lung for carbon monoxide	Independent Liver Safety Data Review Board	Pulmonary veno-occlusive disease
• d/sSAP	• IRB	• PVR
Diastolic/systolic systemic arterial pressure	Institutional Review Board	Pulmonary vascular resistance
• ECG	• IRT	• RHC
Electrocardiogram		Right heart catheterization
• eCRF		
Electronic case report form		

- SAC

Statistical Analysis Center

- SAE

Serious adverse event

- SAP

Statistical Analysis Plan

- SBP

Systolic blood pressure

- SC

Steering Committee

- SD

Standard deviation

- SIV

Site initiation visit

- SmPC

Summary of Product
Characteristics

- SOC

System organ class

- SOP

Standard Operating
Procedure

- SUSAR

Suspected unexpected
serious adverse reaction

- ULN

Upper limit of the normal
range

- USPI

US package insert

- VO2

Volume of oxygen uptake

- WHO

World Health Organization

- WU

Wood Units

PROTOCOL SYNOPSIS

Title:	Efficacy and safety of initial triple versus initial dual oral therapy in adults with newly diagnosed pulmonary arterial hypertension (PAH): a multicenter, double-blind, randomized, placebo-controlled, parallel-group, Phase 3b study.
Design:	Phase 3b, Prospective, multicenter, double-blind, randomized (1:1), placebo-controlled, parallel-group comparison of: - Arm A (Initial Triple Oral Therapy): ERA-01 + PDE5i-01 + IPRA-01 (prostacyclin receptor agonist analogue). - Arm B (Initial Dual Oral Therapy): ERA-01 + PDE5i-01 + placebo (matched to IPRA-01).
Randomization & Stratification:	1:1 via IRT/IWRS; stratified by (a) region (North America vs Rest of World) and (b) baseline WHO Functional Class (I/II vs III/IV).
Blinding:	Double-blind for IPRA-01 vs placebo; ERA-01 and PDE5i-01 are open-label but masked in the overall treatment package; outcome assessors blinded. Emergency unblinding via IRT.
Population:	Adults (18 - 75 years) with newly diagnosed, treatment-naïve PAH (diagnosis < 6 months), WHO Group 1, symptomatic, meeting hemodynamic criteria by right-heart catheterization (RHC). Key etiology subgroups allowed: idiopathic, heritable, drug/toxin-induced, PAH associated with connective tissue disease, HIV, or repaired simple systemic-to-pulmonary shunts (≥ 1 year post-repair).
Planned Sample Size:	~240 participants (~120 per arm) from ~85 sites in ~20 countries.
Study Duration per Participant:	Main observation period ~26 weeks (+ safety follow-up), with optional long-term extension visits every 6 months up to 24 months.
Primary Objective/Endpoint:	Change from baseline to Week 26 in pulmonary vascular resistance (PVR; $\text{dyn} \cdot \text{s} \cdot \text{cm}^{-5}$) measured by RHC.

Key Secondary Objectives/Endpoints:	- Change from baseline to Week 26 in 6-minute walk distance (6MWD) and Borg Dyspnea Index. - Change in WHO Functional Class. - Changes in cardiopulmonary hemodynamics other than PVR (e.g., mPAP, RAP, CI) at Week 26. - Biomarkers of right ventricular function/strain (e.g., NT-proBNP). - Time to first disease-progression event (composite). - Safety/tolerability: adverse events (AEs), serious AEs (SAEs), labs, vitals, pregnancy testing.
Exploratory Objectives:	- Additional disease-severity metrics and health resource use. - Subgroup analyses by etiology and baseline risk.
Visit Schedule Overview (per-patient):	- Screening (Day –28 to Day –1), Day 1 (Randomization), Day 8, Day 15, Weekly phone calls during IPRA-01 uptitration until Week 12, Week 12 visit, Week 26 visit (includes repeat RHC), then Month 12/18/24... visits (extension). - End-of-study (EOS) visit ~30 days after last IP dose or early termination; additional targeted follow-up for certain lab signals (e.g., elevated liver enzymes).

BACKGROUND & RATIONALE

Pulmonary arterial hypertension (PAH) is a progressive disorder characterized by increased PVR leading to right heart failure. Combining an endothelin receptor antagonist (ERA) with a phosphodiesterase type-5 inhibitor (PDE5i) is standard initial therapy for many newly diagnosed patients at low/intermediate risk. Adding a prostacyclin-pathway agent (IP receptor agonist, IPRA-01) at initiation may improve hemodynamics and clinical outcomes. This study compares initial triple vs dual oral therapy in treatment-naïve PAH.

A phase 2 study has investigated initial triple combination therapy, showing dramatic improvements in all relevant variables such as WHO FC, 6MWD, mPAP, cardiac index, and PVR.

The triple therapy data from the previous study provides a rationale for the present study. The purpose is to investigate whether an initial triple oral treatment regimen combining adds significant efficacy benefit while being safe and well tolerated, as compared to an initial dual oral treatment regimen.

Given the extensive and long-term controlled efficacy and safety data available with all 3 study treatments and the careful follow-up of subjects mandated by this protocol, the benefit/risk assessment supports the current study.

Interventions:

- **ERA-01 (oral):** once daily fixed dose (e.g., 10 mg qd).
- **PDE5i-01 (oral):** 20 mg qd for 7 days then 40 mg qd as tolerated.
- **IPRA-01 (oral):** initiated Day 15; individualized weekly uptitration (e.g., in 200-mcg BID increments) to highest tolerated dose by Week 12; stable through Week

STUDY OBJECTIVES & ENDPOINTS

Primary Objective:

Compare initial triple vs dual oral therapy with respect to PVR reduction.

Primary Endpoint:

Absolute change in PVR from baseline to Week 26 by RHC.

Key Secondary Endpoints:

- Change in mPAP, RAP, cardiac index at Week 26.
- Change in 6MWD and Borg score at Week 26.
- Change in WHO FC at Week 26.
- NT-proBNP change from baseline.
- Time to first disease-progression event through Week 26 (composite: death, lung transplant, hospitalization for PAH, initiation of parenteral prostacyclin, or sustained clinically significant worsening defined in the SAP).

Safety/tolerability endpoints:

Incidence of AEs/SAEs, labs, vitals.

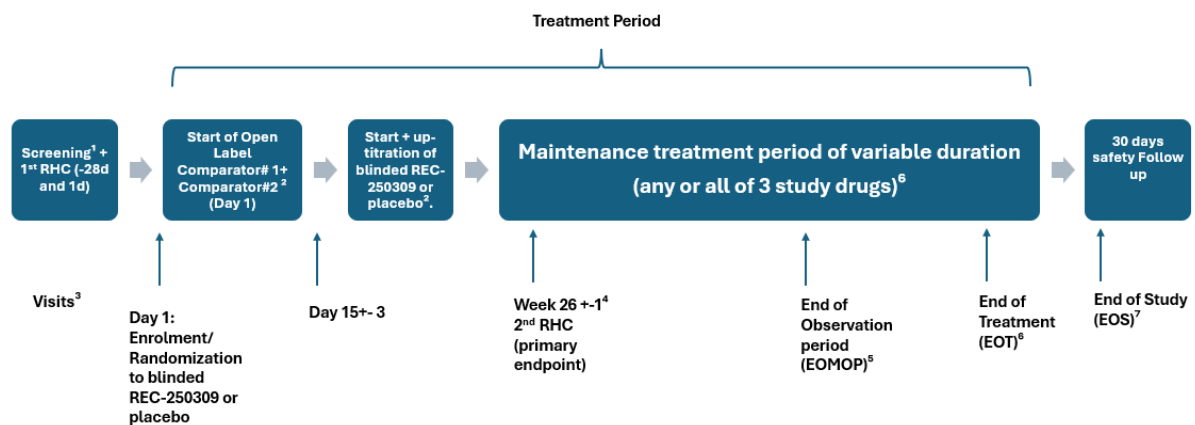
Exploratory endpoints

Additional disease-severity indices and health economics outcomes.

Subgroup analyses (etiology; baseline risk strata).

STUDY DESIGN & PLAN

Prospective, randomized, double-blind (IPRA-01 vs placebo), placebo-controlled, parallel-group.



Randomization

1:1 via IRT; block size concealed; stratified by region and WHO FC.

Blinding

Participants, investigators, assessors blinded to IPRA-01 vs placebo; emergency unblinding via IRT.

Sample Size

~240 participants (~120 per arm).

Sites & Countries

Approximately 80 – 90 sites across ~20 countries.

Oversight

Independent Data Monitoring Committee (IDMC/DSMB): periodic review of unblinded safety; stopping rules defined in charter.

Endpoint assessments

Standardized procedures for RHC (catheter position, zeroing, measurement at end-expiration), 6MWT per guidelines.

Duration

Per-subject: Screening up to 28 days; treatment to Week 26; EOS ~30 days post-last dose; optional extension visits every 6 months (up to Month 24 or site close-out).

STUDY POPULATION

Adults with symptomatic WHO Group 1 PAH, **newly diagnosed** (<6 months), **treatment-naïve**, hemodynamically confirmed. ~240 participants (~120 per arm).

Inclusion Criteria

1. Signed ICF before any study-specific procedures.
2. Age 18-75 years at screening.
3. PAH diagnosis <6 months before Day 1.
4. Screening/baseline RHC (–28 to Day 1) meeting **all**:
5. mPAP ≥ 25 mmHg; PAWP/LVEDP ≤ 15 mmHg;
6. PVR ≥ 6 Wood Units; negative acute vasoreactivity (mandatory for idiopathic/heritable/ drug-induced PAH).
7. Etiology: idiopathic, heritable, drug/toxin-induced, PAH-CTD, PAH-HIV, or repaired simple shunts (≥ 1 year).
8. 6MWD ≥ 50 m at screening.
9. Women of childbearing potential (WOCBP): negative pregnancy tests (screening serum; Day 1 urine), monthly testing, and effective contraception through 1 month after last dose,

Exclusion Criteria

10. Any prior PAH-specific therapy before Day 1 (except for short-acting use during vasoreactivity testing; remote intermittent use for non-PAH indications allowed if >6 months prior). Recent structured exercise-based cardiopulmonary rehab (planned or started ≤ 12 weeks before Day 1).
11. BMI > 40 kg/m².
12. Significant lung disease, left-heart disease, chronic atrial fibrillation, or multiple risk factors for HFpEF suggesting non-PAH PH.
13. Significant hepatic or renal impairment (thresholds per lab manual).
14. Pregnant or breastfeeding.
15. Full, study-ready inclusion/exclusion criteria are provided to enable pre-screening and screening decisions in the workbook.

Special Populations

WOCBP: contraception & monthly pregnancy testing.

Contraception Methods: acceptable methods and durations.

Option 1	Option 2	Option 3	Option 4
One method from this list: Standard IUD (Copper T380A IUD) Intrauterine system (LNg 20IUS: progesterone IUS) Progesterone implant Tubal sterilization	One method from this list: Estrogen and progesterone oral contraceptives ("the pill") Estrogen and progesterone transdermal patch Vaginal ring Progesterone injection PLUS one method from this list: Male condom Diaphragm with spermicide Cervical cap with spermicide	One method from this list: Diaphragm with spermicide Cervical cap with spermicide PLUS one method from this list: Male condom	One method from this list: Partner's vasectomy PLUS one method from this list: Male condom Diaphragm with spermicide Cervical cap with spermicide Estrogen and progesterone oral contraceptives ("the pill") Estrogen and progesterone transdermal patch Vaginal ring Progesterone injection

IUD, intrauterine device; IUS, intrauterine system.

TREATMENTS

ERA-01: Oral tablet, 10 mg once daily.

PDE5i-01: Oral tablet, 20 mg once daily for 7 days then 40 mg once daily if tolerated; dose reductions allowed for AEs.

IPRA-01 / Placebo: Oral tablet, initiated Day 15; individualized weekly up titration (e.g., 200-mcg BID increments) until Week 12; maintain highest tolerated dose to Week 26. Temporary down-titrations for tolerability permitted.

Day 15 to Day 21	200 µg twice daily	(1 tablet twice daily)
Day 22 to Day 28	400 µg twice daily	(2 tablets twice daily)
Day 29 to Day 35	600 µg twice daily	(3 tablets twice daily)
Day 36 to Day 42	800 µg twice daily	(4 tablets twice daily)
Day 43 to Day 49	1000 µg twice daily	(5 tablets twice daily)
Day 50 to Day 56	1200 µg twice daily	(6 tablets twice daily)
Day 57 to Day 63	1400 µg twice daily	(7 tablets twice daily)
Day 64 to Week 12 visit	1600 µg twice daily	(8 tablets twice daily)

Treatment Assignment & Blinding

IRT assigns kits and supports emergency unblinding. Pharmacy dispenses per kit list;

subject-level blinding maintained.

Dose Adjustments & Interruptions

Hypotension, headache, GI AEs: step-down titration or temporary interruption per management guide; resume titration when resolved.

Liver enzyme elevation: see Safety labs and LFT follow-up algorithm.

Concomitant Therapy

Initiation of parenteral prostacyclin is a disease-progression event but permitted for safety.

Prohibited medications listed below:

- Any PAH-specific drug other than the 3 study treatments up to EOT, except if used as rescue therapy.
- Organic nitrates
- Strong inducers of CYP3A4
- Strong inhibitors of CYP3A4
- Strong inhibitors of CYP2C8
- Any investigational drug other than the 3 study treatments.

IP Supply, Accountability, Storage

Supply: Centrally packaged, labeled for blinded use; shipped temperature-controlled per vendor SOPs.

Storage: Ambient locked storage with temperature monitoring and access control; separate area for returns/quarantine.

Accountability: Pharmacy maintains receipt/dispense/return/destruction logs; investigational product reconciliation at each monitoring visit.

STUDY ASSESSMENTS

Screening/Baseline Assessments

- Eligibility check (I/E criteria).
- Demographics; medical history; prior therapies.
- RHC (–28 to Day 1).
- Physical exam; vitals (BP/HR); WHO FC; 6MWD; Borg dyspnea.
- Central labs (general safety; serum pregnancy for WOCPB); biomarkers.

Efficacy Assessments

- **RHC:** standardized zeroing/position; measurements include mPAP, PAWP, RAP, CI, PVR (calculated).

- **6MWT:** per guideline (standardized course; same time of day when possible; pre-test vitals).
- **WHO FC:** trained clinician assessment.
- Biomarkers: NT-proBNP.

Safety Assessments

- **AEs/SAEs:** collected continuously from consent until EOS; SAE reporting within 24 hours to Sponsor Pharmacovigilance.
- Contraception & Pregnancy Testing: WOCBP monthly through EOS.
- Vitals/Weight/Physical Exam: at scheduled visits.
- **Concomitant Medications:** recorded at each contact.
- **Laboratory Tests:** Central chemistry/hematology monthly during titration; thereafter per visit schedule.

Liver Safety Follow-up

If **ALT or AST >3× ULN** (confirmed): repeat tests within 48 – 72 h and add bilirubin, ALP; evaluate symptoms. If **ALT or AST >5× ULN** or **ALT/AST >3× ULN with bilirubin >2× ULN** (no cholestasis): **hold** study drugs; increase monitoring frequency; consider Hy’ s law evaluation; manage per medical judgment; report as SAE if criteria met. Resume dosing when values improve to <3× ULN and subject asymptomatic.

Schedule of Assessments

Visit Windows: Day 8/15 (±2 d); Week 12 (±7 d); Week 26 (±14 d); EOS (±7 d).

Procedure/Period	Screening ⁶	Treatment							EOT ⁵	Follow Up	
	Day -28 to Day 1	Day 1	Day 8 ¹	Day 15	Weekly Phone Calls ²	Week 12	Week 26	Month 12, 18, 24, 30, etc		EOS ¹⁵	Unsch ³
Informed consent ¹⁰	X										
Eligibility and demographics	X										
Medical History	X										
Concomitant Medications	X	X		X		X	X	X	X	X	X
Right heart Catheterization	X ¹⁰						X				X
Physical Examination	X			X		X	X	X	X		X
Vital Signs	X	X ⁷		X		X	X	X	X		X
WHO FC ¹²		X ⁷				X	X	X	X		X
6 WWD ¹² , Borg Dysp. Index	X	X ⁷				X	X	X	X		X
Lab tests ¹⁴	X ¹³	X ⁷				X	X	X	X		X
Liver and Haemoglobin tests ⁹	X ¹³	X ⁷	Monthly (every 30days)							X	
Pregnancy Test ⁸	X ⁴	X	Monthly (every 30days)							X	X
Biomarkers	X						X				
IP Dispensing		X		X		X	X	X			
IP Return						X	X	X	X		X
AE/SAE ¹¹	X	X	X	X	X	X	X	X	X	X	X

¹ Visit or phone call.

² Weekly phone calls during the up-titration phase of double-blind study treatment.

3 Unscheduled visits may be performed at any time during the study and may include any of the indicated assessments, based on investigator judgment.

4 Reliable contraception must be started at least 11 days prior to Day 1.

5 EOT, EOS: See definitions under 'Periods' above.

6 It is permitted to re-screen subjects once, if the reason for non-eligibility was transient (e.g., abnormal laboratory test, insufficient wash-out period of a forbidden medication). All screening assessments should then be repeated at the time of re-screening.

7 Only needed if screening assessment > 7 days before Day 1.

8 In women of childbearing potential only. Serum pregnancy tests at Screening, Week 12, Week 26, monthly (± 1 week) throughout the study, at EOT, and if pregnancy is suspected at any time during the study. Urine pregnancy tests at the Day 1 visit and if a monthly serum test is missed. At EOS either serum or urine pregnancy testing.

9 Monthly (± 1 week) central laboratory testing of liver aminotransferases, total and direct bilirubin, and hemoglobin is mandatory throughout the study. For monthly testing, blood drawing is performed either at the study site, at a satellite laboratory close to where the subject lives, or by a phlebotomy service at the subject's home.

10 Signed informed consent required before any study-mandated procedure. However, RHC data obtained at the study site within 28 days prior to Day 1 but before the study (i.e., before signed informed consent) are acceptable.

11 All AEs and SAEs that occur after signing the Informed Consent Form and up to 30 days after discontinuation of the last of the 3 study treatments must be reported.

12 If any post-baseline decrease in 6MWD by > 15% from the highest 6MWD obtained at or after screening is observed, accompanied by WHO FC III or IV, then both the 6MWD and WHO FC must be reassessed at an unscheduled visit after 1 – 21 days.

13 As an alternative to central laboratory tests, eligibility of subjects at Screening may be determined using local laboratory tests as long as the central laboratory kit is used in parallel. All local laboratory data including the normal ranges must be entered on dedicated eCRF pages.

14 Laboratory tests general: Hematology, clinical chemistry including liver and hemoglobin tests, coagulation tests, and NT-proBNP.

6MWD = 6-minute walk distance; AE = adverse event; BP = blood pressure; eCRF = electronic case report form; EOS = end of study; EOT = end of treatment; FC = functional class; HR = heart rate; NT-proBNP = N-terminal prohormone of brain natriuretic peptide; RHC = right heart catheterization; SAE = serious adverse event; WHO = World Health Organization.

EARLY DISCONTINUATION

Early Discontinuation Triggers: participant withdrawal, pregnancy, unacceptable toxicity, protocol-defined liver enzyme thresholds, eligibility violation discovered post-randomization (handled per SAP), loss to follow-up.

Requirements if Early Discontinue: perform EOT/EOS safety visit; RHC at Week 26 not required if off study unless medically justified.

SAFETY REPORTING & DEFINITIONS

AE: any untoward medical occurrence after consent (regardless of causality).

SAE: results in death, life-threatening event, inpatient hospitalization or prolongation, significant disability/incapacity, congenital anomaly, or other important medical events.

SUSAR: suspected, unexpected serious adverse reaction (blinded assessment per PV plan).

Timelines: SAEs reported within 24 hours of awareness; pregnancies within 24 hours; follow through to outcome.

DATA MANAGEMENT & QUALITY

EDC: role-based access; real-time edit checks; audit trail; query management; SDV/RBM per monitoring plan.

Source Documentation: contemporaneous, attributable, legible, original, accurate, complete (ALCOA+); certified copies permitted via validated process.

Essential Documents: maintained in Investigator Site File (ISF) per ICH E6 Section 8; archival per local law.

Risk-Based Monitoring: focus on critical data/processes (eligibility; IP exposure; primary/secondary endpoints; safety reporting; pregnancy testing; visit adherence).

PROCEDURES AND GOOD CLINICAL PRACTICE

Sponsor and the investigators will ensure that the study is conducted in full compliance with the latest versions of ICH-GCP Guidelines, the principles of the “Declaration of Helsinki” , and with the laws and regulations of the country in which the research is conducted.

Informed consent

It is the responsibility of the investigator/delegate to obtain informed consent according to ICH-GCP guidelines and local regulations from each individual participating in this study and/or legal representative. The investigator/delegate must explain to subjects that they are completely free to refuse to enter the study, or to withdraw from it at any time for any reason and without having to provide any justification.

The ICF will be provided in each country’ s local language(s).

Site staff authorized to participate to the consent process and/or to obtain consent from the subject

Protocol adherence/compliance

The investigator must conduct the study in compliance with the approved version of the protocol and must not implement any deviation/change from the protocol, except when deviation is necessary to eliminate an immediate hazard to the subject.

If a protocol deviation occurs, the investigator/delegate will inform [Sponsor Name] or its representative, in a timely manner. The investigator/delegate must document and explain any deviation from the approved protocol. Deviations considered to be a violation of GCP must be reported to the IRB/IEC and regulatory authorities according to Sponsor or (overruling) local requirements.

Essential documents and retention of documents

The investigator/delegate must maintain adequate records necessary for the reconstruction and evaluation of the study. A number of attributes are considered of universal importance to source data and the records that hold those data. These include that the data and records are accurate, legible, contemporaneous, original (or certified copy), attributable, complete, consistent, enduring, and available when needed. These records are to be classified into two different categories of documents: ISF and subject clinical source documents. These records must be kept by the investigator for as long as is necessary to comply with Sponsor’ s requirements (i.e., as specified in the clinical study agreement), and national and/or international regulations, whichever would be the longest period. If the investigator cannot guarantee this archiving requirement at the site for any or all of the documents, special arrangements, respecting the data confidentiality, must be made between the investigator and Sponsor to store these documents outside the site, so that they can be retrieved in case of a regulatory inspection. No study document should be destroyed without prior written approval from Sponsor.

APPENDIX A – AMENDMENT HISTORY

Amendment 1:

- Clarified BMI exclusion threshold;
- added monthly biomarker timepoint at Week 12;
- widened Week 26 window from ± 10 to ± 14 days;

- updated contraception to include additional LARC methods.

OPERATIONAL LOGISTICS

(AS IDENTIFIED IN STUDY MANUALS PROVIDED BY SPONSOR)

Facilities & Equipment

- 6MWT corridor (≥ 30 m), stopwatch, pulse oximeter, BP device, calibrated scale/stadiometer.
- RHC capability or referral pathway; standardized measurement methods.
- Pharmacy: locked, temperature-monitored storage; accountability system; kit receipt/dispenses/ return logs.
- Lab: centrifuge; ambient/4° C temporary storage; –20° C freezer for aliquots (central lab); shipping supplies; courier pick-ups.

Systems & Accesses

- EDC
- IRT/IWRS
- Central Lab Portal
- PV Safety Portal (for SAE reporting)
- e-mail or e-portal for ethics submissions
- CTMS (if applicable)
- ePRO
- training LMS for vendor modules.

Vendors/Services

- Central lab (kits, requisitions, shipments).
- IRT/IWRS (randomization, kit management, emergency unblinding).
- Courier (ambient) for lab shipments.
- Home health/phlebotomy (optional) for difficult-to-travel participants.

Roles & Responsibilities

- **PI/Sub-I:** medical oversight, eligibility confirmation, safety decisions, RHC oversight, EDC sign-off.
- **Study Coordinator/Research Nurse:** visit scheduling, consent facilitation, data entry, sample handling, IP accountability with pharmacy, logistics.

- **Pharmacist/Designee:** receipt/storage/dispense/returns; compliance counseling; reconciliation.
- **Lab Technician/Designee:** sample processing, packing, shipping; chain of custody.
- **Rater (6MWT/WHO FC):** perform standardized assessments.
- **Data Manager/Monitor:** data quality, SDV, RBM triggers.

Key Workflows

Pre-screening → Consent → Screening/RHC → Randomization → IP dispensing → Titration (with weekly calls) → Week 12 → Week 26 (RHC) → EOS.

Critical handoffs: eligibility confirmation; IP titration decisions; lab alerts (e.g., LFTs); SAE reporting; pregnancy alerts.